

Storyboarding

Rapid CD Process	Lightning Fast	Lightning Fast +	Focus Rapid
Storyboards			

Storyboards take the broad view of the vision, and work out in detail how a person accomplish specific tasks in the new design. You need to figure out the details of work will be done in the new system to ensure that you don't break the users' existing work practice. Therefore, ~~storyboards are guided~~ by and checked against the ~~dated~~ sequences for tasks and intents that need to be supported.

Too often when people design, they break work practice because they jump from big idea to low-level user interface and implementation design. As soon as design start focusing on technology, technology and its problems become their central concern. Storyboarding, guided by the sequence models and affinity data, ensure the team does not overlook any intents and steps that are critical to the work. Storyboarding lets you work out the details of how to support the sequence within the system, whether you support it by making sure users can still accomplish an activity the old way, or by introducing new technology to better achieve the intent.

The vision is a natural prioritizer. You need not storyboard every sequence. Instead, identify which sequences are relevant to the vision, and storyboard those. The sequence models should represent the core tasks to achieve the work within project scope. Supporting those sequences will be enough for a next product release.

Storyboards are equivalent to future scenarios, to-be use cases in RUP, and the basic user stories in the XP process (see the box, **Why storyboard: using drawings instead**

text-based use cases). See Chapter 15 for a more in-depth discussion of how storyboards are related to other methodologies. This chapter illustrates how storyboards are created.

Definition

Storyboards are like freeze-frame movies of the new work practice. Like storyboarding in film, the team draws step-by-step pictures of how people will work in their new world. Storyboards include manual steps, rough user interface components, system activity and automation, and even documentation use. Storyboards act like high-level use cases.

The role of storyboarding in system design is discussed further in Chapter 13, “Design from Data,” pg. 287, in *Contextual Design: Defining Customer-Centered Systems*.

Key concepts

Low-level or mini vision. The storyboarding process starts with a low-level vision. This is a similar process to the visioning process described in Chapter 11. The low-level vision works out the details of a portion of the high-level consolidated vision. The low-level vision is tied to a task defined by a consolidated sequence—working out changes to every step and designed to support sequence intents while overcoming user issues. The low-level vision lets the team explore alternatives and have discussions grounded in their drawings in preparation for storyboarding.

Process for storyboarding

- ☐ Preparation
 - ☐ Prepare the room, data, and supplies
 - ☐ Gather people to storyboard
 - ☐ Choose where to start
- ☐ Review how to storyboard
- ☐ Create a low-level vision
 - ☐ Walk the consolidated vision
 - ☐ Collect issues from the affinity and consolidated sequence model that are related specifically to what is going to be storyboarded
 - ☐ Create a low-level vision for one task to be storyboarded
 - ☐ Identify pluses and minuses for the low-level vision
- ☐ Create the storyboards
 - ☐ Draw the storyboard frames
 - ☐ Check the storyboard against the consolidated sequence
- ☐ Share the storyboard with stakeholders or members of a larger team
- ☐ Refine the storyboard
- ☐ Go on to the next storyboard



Why storyboard? Using drawings instead of text-based use cases

Storyboards are equivalent to future scenarios or high-level use case instances and may be represented in standard use case document structure and Unified Modeling Language notation. We choose to represent the redesigned work in frame-by-frame drawings. They include manual steps, rough user interface components showing function, system activity, automation, and even documentation use.

By working with multiple teams we have learned that the best system design comes from this type of visual representation of the to-be models of work. Text versions of a scenario or simple user interface drawings alone fail to consider all the dimensions of the new work practice. Use case and object modelers are trained to use the text-intensive use case characterization of the redesigned work. We find that this ultimately focuses the team too much on system activities and business rules and not enough on necessary human processes and user interface function. But user interface designers who are concerned primarily with the interaction design tend to overlook the system steps and over-focus on UI detail. They get stuck prematurely in detailed discussions of layout that will become irrelevant when the implications of multiple scenarios of use are worked through.

One key principle of Contextual Design is to use the right representational form for the design conversation the team is trying to have. Even though it might be possible to represent the storyboard concepts with UML models, use cases, stories, a series of UI drawings or even a high-level business process drawing—none of these individual representations encourages the team to think about all these factors simultaneously. Storyboards by their very nature ensure synthetic sequential thinking and thereby a more complete design for the user, especially when guided both by the vision and the consolidated sequences.

Storyboards also provide a conceptual common denominator for design discussions with users; it is much more likely that a user will grasp the metaphor of a storyboard (e.g., as used in the creation of animations, comic books, and movies) over more abstract modeling techniques. When developing in-house systems we photograph our storyboard frames and display them for user groups and stakeholders. Then we run a “share” session, collecting their issues. In this way the team creates buy-in with future users and gets additional feedback that can be rolled into the next version of the storyboard.

See Chapter 15 for a description of how CD is used with other development methodologies.

Prepare the room, data, and supplies

In order to create the storyboards you will need to hang up the consolidated visions and sequences in your team room. You will also need flipcharts for drawing low-level visions and 5½x8½ sheets of paper for your storyboard frames. Bring removable tape to hang the storyboard frames onto flipcharts for review.

Gather people to storyboard

It is best to create storyboards in pairs. As with other parts of Contextual Design, you do not want to storyboard by yourself—with two people you can bounce ideas off each other. If you have more people on your team or want to include others in the process you can create several storyboards in parallel. Be sure that one core team member works on every storyboard to ensure continuity.

If you have core team members who work all the time and adjunct team members who work part time be sure that the same people are always responsible for the same storyboard. Moving people on and off a storyboard slows everyone down because the work has to be reviewed and agreed to over and over. Assign two people to each storyboard and then run a formal share session later.

Choose where to start

Before jumping into storyboarding you need to pick a place to start. Your sequences represent the key tasks for your project. Select the sequences you will storyboard starting with the work that is core to the system.

If you are storyboarding in parallel with stakeholders or a larger team, assign two people to each sequence to storyboard. Ensure that the sequences do not have overlapping activities—if so, indicate what team will address the overlapping piece. If you have very long sequences consider breaking them up to give to teams working in parallel. If you are a two-person team, simply start with the core storyboards.

Review your consolidated vision to see if it implies tasks that do not correspond to your sequences. You may want to get additional data to guide the storyboarding—field interviews focused on collecting data for one task are very quick. Even having one real instance to guide the storyboarding will be quicker and more productive than making up the storyboard with no data.

We have seen teams spin their wheels and argue when they have no clear idea of how an activity is accomplished. The vision alone is not enough of a guide, particularly for a complicated or unfamiliar task.

Review the vision for different cases or conditions. Your consolidated sequence will also have different branches and strategies—storyboard each of these as well. Sometimes new process design plans within a business will imply that the same activity will be storyboarded under different assumptions. Make a list of all cases implied by the vision, the consolidated sequence, and any process plans, and storyboard these cases.

Review how to storyboard

Be sure that everyone participating understands the following:

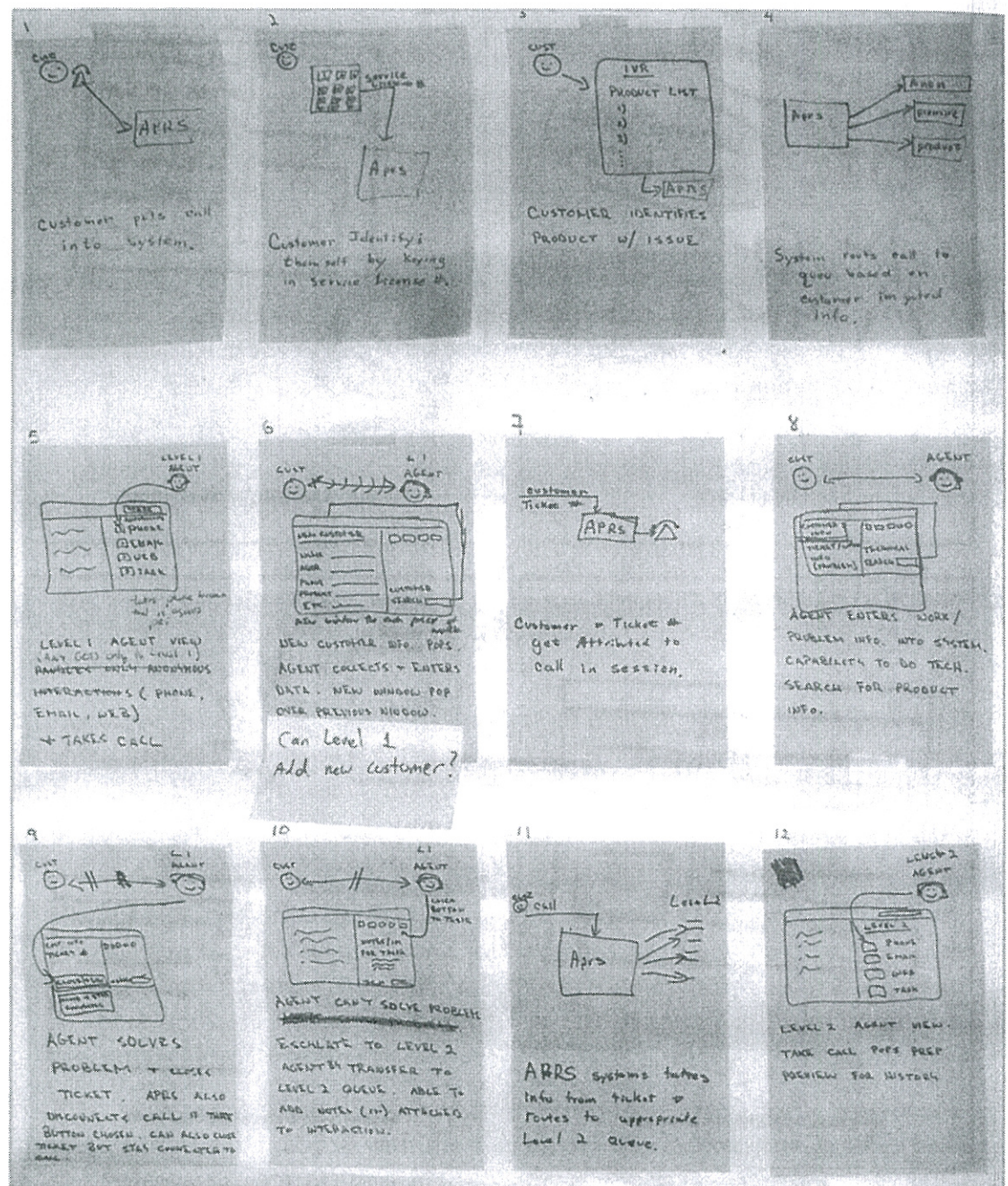
- The storyboard works out in more detail the new work practice described in the vision. It is made up of both pictures and words representing the complete story of the work, not just interactions with the product.
- A storyboard is a set of hand-drawn pictures that let people think concretely about how the new work practice will work. It includes manual steps, interaction with the new system's user interface, system automation, business rules, and interaction with other systems and products. See Figure 12-1 for examples of different types of storyboard frames.
- Storyboards are guided by user data. Each storyboard will be associated with an existing consolidated sequence and will take into consideration other user data gathered from the affinity and artifacts. When the storyboard is done you want to check it against the sequence to be sure that users can still accomplish the task and achieve their intents.
- Each storyboard frame is one step in the story. Each step in the storyboard represents one interaction with the system, one interaction with another person, one manual step, or one behind-the-scenes step where the system responds.

Describe the order of the process:

1. Collect the data relevant to your sequence.
2. Create a low-level vision of your sequence accounting for each activity, intent, and step while staying within the constraints of the vision and the agreed to technology platform.
3. Do pluses and minuses on the low-level vision and fix the minuses.
4. Create the storyboard frames. Be sure to put only one interaction or step within each frame.

See Figure 12-2 for an example of a completed storyboard.

Figure 12-2:
An example of an actual storyboard. Note the different types of views on the work—process, UI, and human interaction. The Post-it® notes are issues that were gathered after a storyboard share.



- Just like a movie a storyboard can have flashbacks and flash forwards to deal with strategies and cases—it doesn't have to be one straight path.
- A storyboard should have enough detail and text for someone who did not create it to read and understand the proposed process. You also want others not on the team to be able to understand them. So in a business, your employees or business customers can review the storyboard to see what the new process will be like.

Types of Storyboard Frames

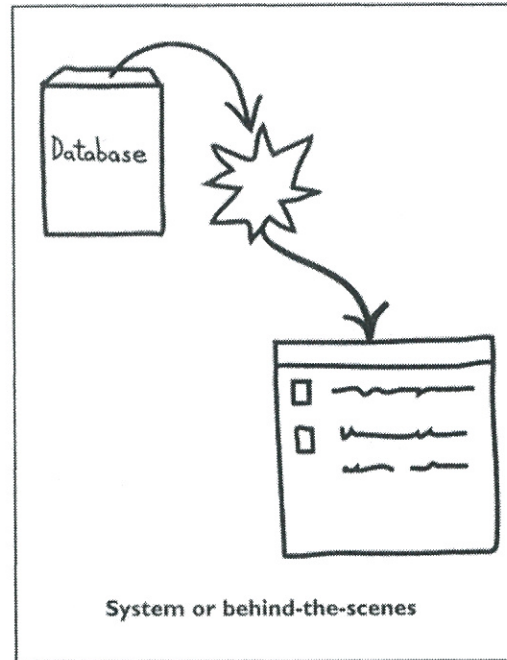
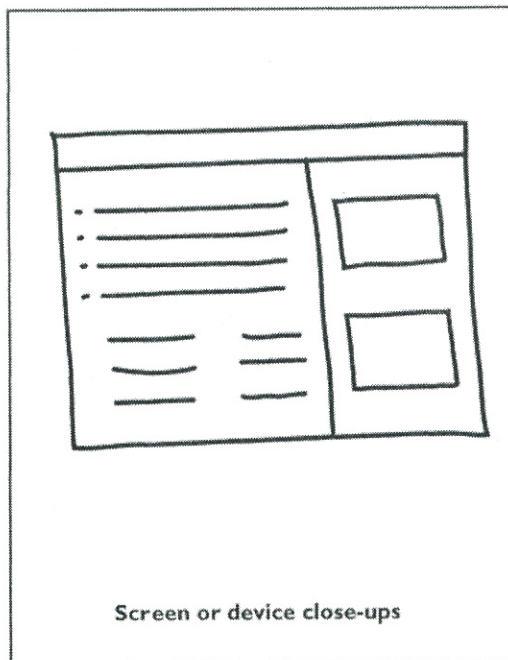
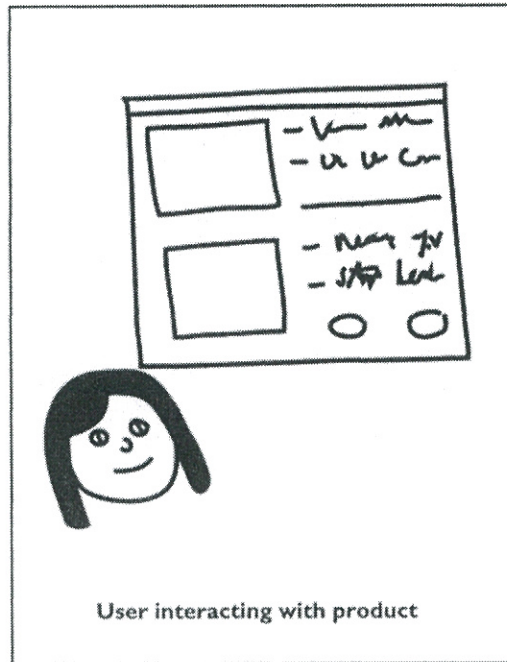
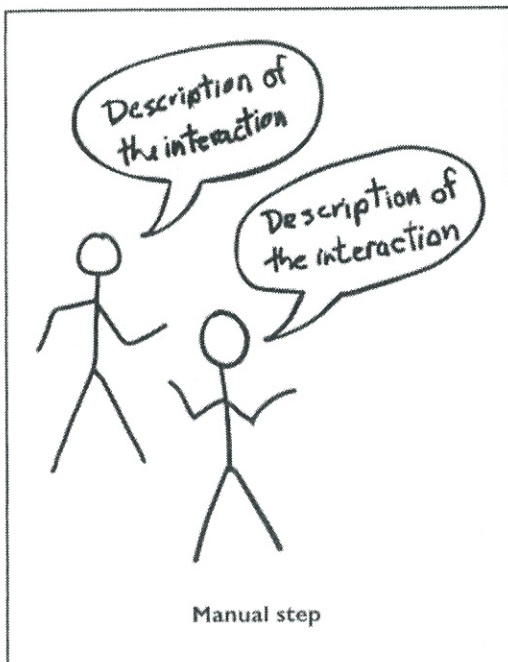


Figure 12-1:
Examples of different types of storyboard frames.

Draw the storyboard

Now you are ready to capture the storyboard according to the new work practice. Remember, each frame represents one step in the story. If there are branches to a step, let the storyboard branch and number each frame; for example, 1A, 1B, 1C.

Once you are finished, tape the frames to a flipchart page so the storyboard can easily be seen and used. Write a title for the storyboard, and number each of the frames.

When drawing the storyboard frames remember the following:

- Draw rough user interfaces annotated with usage.
- Describe manual actions—draw pictures where needed for clarity.
- Sketch documentation used.
- Draw cartoons of people interacting for interpersonal steps.
- If the system will take over a step draw what the system does and describe any data it relies on.
- When the user does a series of actions with the system, draw a separate step for each action with rough layout of the user interface.
- Annotate with role names to indicate the players performing the tasks.
- Annotate with any business rules, implementation notes, constraints, hardware or software expectations, etc.
- Include words to explain what is happening, state assumptions, and anything else where a picture is not sufficient.
- Assume that you are creating “documentation” of the process in the storyboards so that someone that wasn’t in the design meeting could read the storyboard and understand what is happening.

Drawing the storyboard frames should take about one hour if you think things through the steps in the low-level vision.

Example: eChalk

Figure 12-3 shows a flipchart page with 15 frames of eChalk’s total storyboard for their Online Classroom vision. Two two-person subteams worked on the storyboards, and then the storyboards were brought together and shared to be sure the overall story hung together.



They did a total of 34 frames to represent the following tasks:

- Creating a class calendar
- Making and receiving assignments
- Creating class events
- Taking attendance
- Entering grades
- Communicating with parents

In a storyboard you want to show the process, the exact underlying business rules or automation desired, the content of the interface and an initial layout, the collaboration expectations, the reliance on other technology, back-end systems that need to be accessed, and any other elements relevant to getting the work of the task complete. You are working out the functions needed to support the work but within the context of the whole work task and preliminary concepts of how to chunk access to function in the user interface.

Creating a low-level vision should take 1 to 1.5 hours per task or central case. More complicated tasks with many conditions may take longer.

***Caution:** When feeling pressed for time, don't be tempted to skip the low-level vision and go directly to creating the storyboard frames. It actually takes longer to go straight to the storyboard because you haven't taken the time to think it through. The low-level vision does that for you, and then creating the storyboard frames is essentially a transfer process from the low-level vision to the frames.*

Identify pluses and minuses for the low-level vision

Once you have your possible approaches, identify pluses and minuses just like you did after visioning. Do this even if you had only one approach because it causes you to step back and think critically about your solution.

Generate design ideas for the minuses and fix them in the low-level vision. If you came up with alternate methods of doing the same task, run pluses and minuses on the alternates in order to choose which you are going to use.

If you have competing possibilities you must choose one to storyboard. If you can't agree on the storyboard direction after pluses and minuses, pick the more radical solution (see the box, **How to choose between two ideas**).

Keep track of your alternatives—you can always fall back on them if the user doesn't like the design you choose to present. We call these alternatives test cases. They help define your focus for paper prototype testing (see Chapter 13), so keep track of them.



How to choose between two ideas

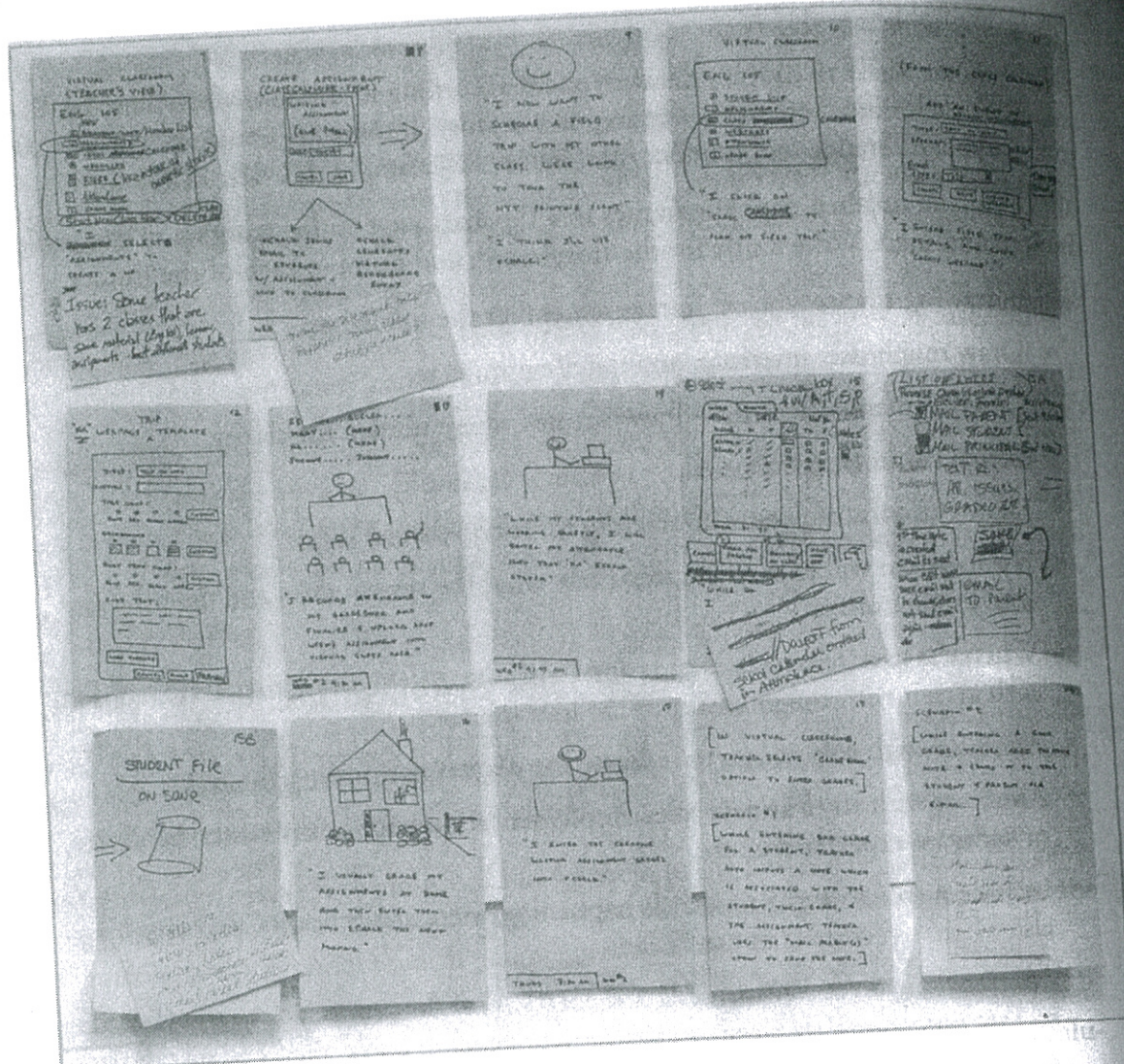
When low-level visioning you may have two alternatives that are equally good. Within the Rapid CD process storyboards will drive development of the paper prototypes, which will then be tested with the user (see Chapter 13)—so you don't need to worry about having picked the “best” solution of the two. But how do you choose between them? A good rule of thumb is to pick the more radical of the choices, the one the user is least likely to expect. Your storyboard is on the way to becoming what you test with users in a paper prototype. If you give the users what they are accustomed to seeing, you won't elicit much of a reaction. However, when it is unexpected, the user will react—either positively or negatively—and you can use that reaction to drive a conversation.

So we recommend picking the solution that challenges the users' habits but better supports their intents. You can always fall back to a more traditional approach after testing.

Then create a list of test cases so you know what you want to test when you are out in the field. Remember, the final arbitrator of the goodness of the design is the user, not you.

Chapter Twelve: Storyboarding

Figure 12-3: Talk storyboard with issues in review session.



Different cases or scenarios were run for some tasks like grading, which had two cases: one for a good grade and another for a poor grade. Classroom assignments also had two cases—the teacher's perspective and the student's perspective.

The eChalk team wanted to be sure they thought through the interactions between teacher and eChalk throughout the day. So they indicated the transition of time during the day as the teacher goes from task to task (i.e., Wed. 8:30 a.m., Wed 9:30 a.m.). Manual steps flesh out the story and ensure that the team is thinking about how the technology fits into the work practice smoothly. System behavior is storyboarded to show when it drives what the user sees onscreen, and when data is stored for future use.

The team is not worried about making neat or beautiful storyboards; they are focused on driving out the details of the work. They have tried to make each storyboard frame have enough detail that someone familiar with the project understands the work practice being represented.

Figures 12-4 through 12-8 show a closer look at some of eChalk's storyboard frames.

Figure 12-4 shows a screen with all the functions in eChalk. The team is using this frame to start the story with the equivalent of an opening scene. This sets the stage for the various functions the teacher will use in the upcoming frames. In this sketch the user selects the New Assignment function to create a new classroom assignment.

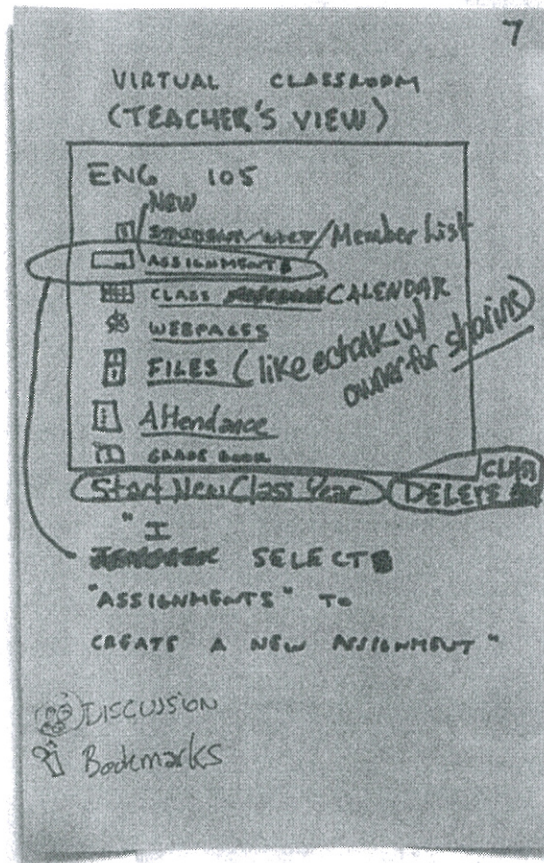


Figure 12-4:
Close-up view of
frame seven from
eChalk, which sets
the context for the
frames that follow.

The top portion of the frame in Figure 12-5 is the user's screen, drawn for reference to explain what is being done behind the scenes by the system. The bottom half of the frame describes the system behavior. Note the time of day as the story of the teacher's day moves along.

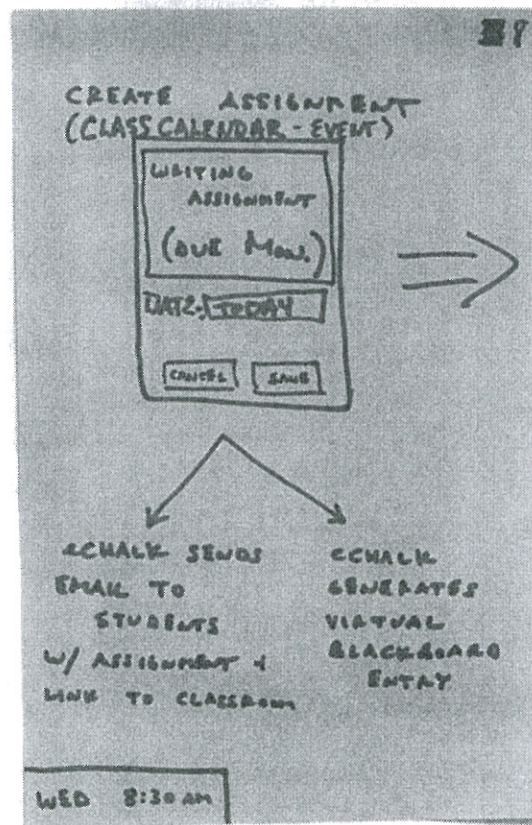
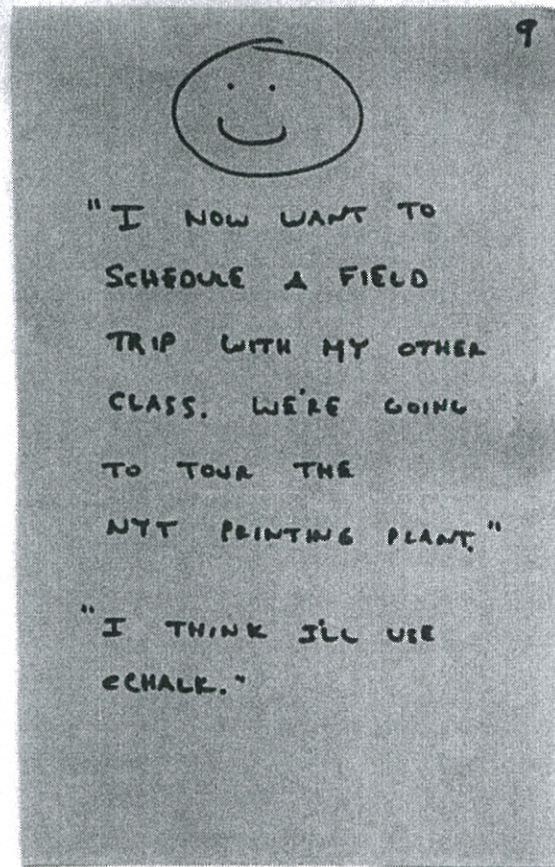


Figure 12-5:
Close-up view of
frame eight from
eChalk that shows
both the UI function
and the system
step.

Figure 12-6:
Close-up of eChalk
frame nine. This
frame uses a manu-
al step to tell the
story and drive out
the work the design
must support.



A manual step makes the transition to the next task to be performed. Figure 12-6 shows the teacher's intent and task. It also shows how the eChalk tool is planned to be hooked into the overall process.

Figure 12-7:
Close-up of story-
board frame eleven.
This frame is using
a real case to reveal
the needed func-
tion.

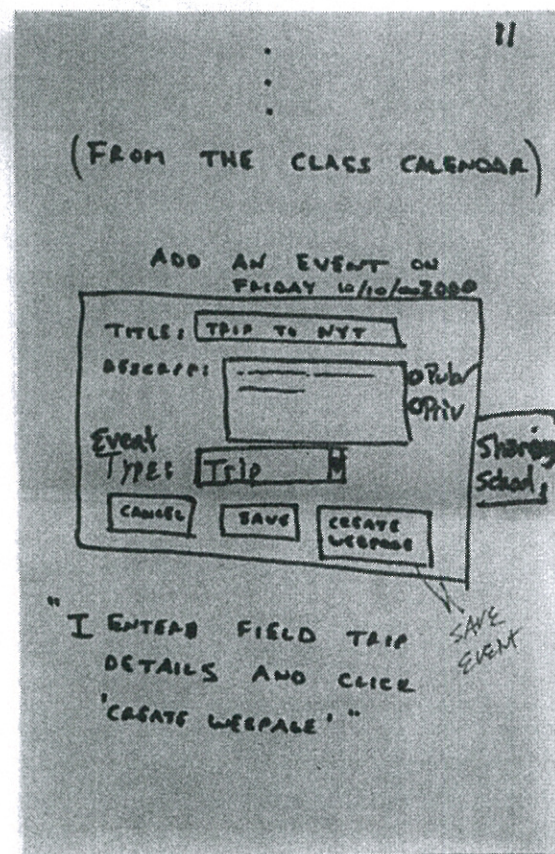


Figure 12-7 shows a screen interaction sketch, with real details about the field trip class event the user is creating. The details come from an actual field trip the team captured in a sequence model.

Figure 12-8 shows a more detailed view of a screen that is presented to the user, with enough UI details to make it clear what functionality needs to be available. Remember that this is not the real UI; the screen sketch is intended to capture only the functions that need to be present to do the task for the user.

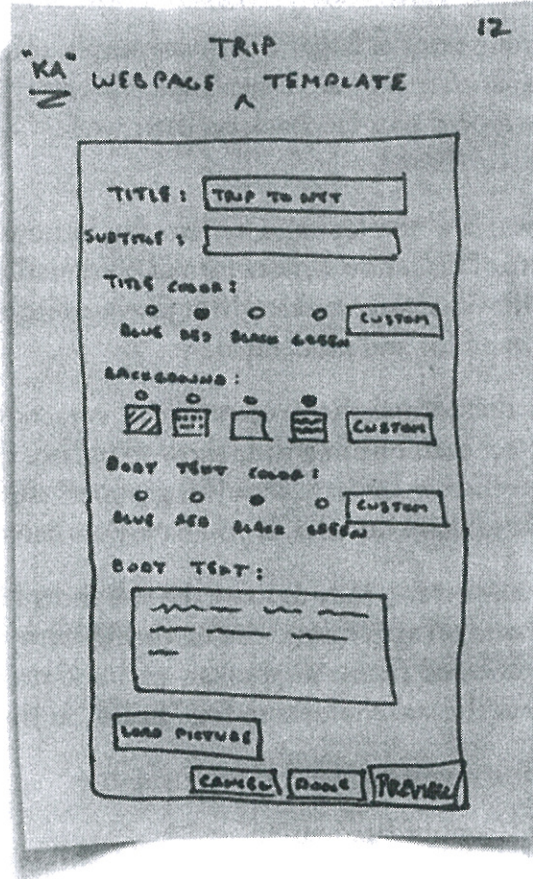


Figure 12-8: Close-up of storyboard frame twelve that represents the needed functions without designing the actual UI.

Tips

Here are some tips about the storyboarding process.

- For storyboarding you are concerned with overall structure and function. *Do not* worry about precise language, icons, or layout. The real UI design will be created later.
- If you get stuck while drawing out the storyboard, go back and low-level vision the part you are stuck on.
- If you seem to keep generating new cases as you work out the sequence you have, ask yourself if these are cases that have appeared in the users' work and therefore must be dealt with, or if they are your idea of what might or could happen. Don't worry about logically deduced cases—get the known user cases right first.

Check storyboard against the consolidated sequence

When the storyboard is complete check it against the consolidated sequence models be sure that you have accounted for each step and intent in the sequence. "Accounted for" means that you have thought about how the step or intent will be achieved. It may be that you have decided that certain steps or intents will be supported the same way they are now with no change in your new design. Or you may have introduced new steps, new technology, or automation. This is a last pass quality check to be sure you did not break the work with your design.

Share the storyboard

If your team is larger than two people and you worked in parallel, share your storyboards after they are completed. This storyboard share is a review session to give each storyboard pair feedback on their work. Capture the issues gathered and then rework the storyboard.

If you are developing a system for in-house users, conduct a storyboard share with them. This is now a more formal presentation where you share your ideas on how their work will change and let them provide feedback and express concerns. So it needs to be planned for and moderated.

For this formal share, use a digital camera to photograph the storyboard frames and project each one in a slide show to project the storyboard in a large room. If you have been sharing the data with this group all along, there is no need for a slide show or wall walk to share the data. If you have not shared the findings and your vision do this first.

When you are ready to share the storyboard frames, project each one and capture issues by hand on each frame. You can bring along the paper storyboards and add a Post-it® to the affected frame so you can go back later and redesign to address the issues. This allows the stakeholders to feel “heard” as they publicly see you putting up their issues.

To run any storyboard share, hang the:

- Consolidated vision
- Storyboard
- Sequence that was the basis for the storyboard

For this meeting to be an effective review with your team, all team members need to be present. You don't want to do this over and over and get feedback one piece at a time. So hold one meeting and tell those that can't make it that they are giving up their opportunity for input. One way to manage time for Rapid CD is to be clear about expectations for participation.

For any storyboard share meeting identify a moderator to run the meeting. This will be one team member. The moderator keeps the meeting focused and on pace. He or she will also decide what discussions should be continued or deferred to a later time. The storyboard will be presented by two of the people from the storyboard team: the presenter and the Pen.

To share a storyboard, the presenter:

- Introduces the sequence or task that the storyboard addresses at a high level to orient the viewers.
- Walks each storyboard frame in order, describing what is happening on it.
- Listens for feedback, issues, or design ideas from the participants writing them on Post-its® and posting them to the relevant frame.

As the presenter talks, the Pen annotates the storyboard for anything the presenter says that they forgot to write on the storyboard. This is another informal quality check.

While the presenter is walking through the storyboard, reviewers should be looking for:

- Missing or confusing aspects of the storyboard
- Violations of user work practice as collected in the user data
- Inconsistency with the vision
- Opportunities for more improvement to the work
- Worries about technology or user acceptance
- Violations of planned business process change
- Inconsistencies with the business mission

Team members also can write their issues or design ideas on Post-it® notes and stick them on the storyboard frame they are related to.

Note: Resolution of issues or adoption of design ideas is the work of the storyboard team. The whole group asks questions to understand the design and the design choices, but they do not engage in resolving issues together in the share session. That's the job of the storyboard team when they rework the storyboard.

Rework the storyboard

The storyboard team takes the notes from the share session and decides how to change the storyboard, and then reworks it using the same storyboarding process.

One rework is sufficient. There is no need to reshare the reworked storyboard unless you want to. Again, remember the storyboard is on its way to testing with users through mock-up interviews. Too much rework just wastes time better spent with the user.

Go on to the next storyboard

Once you are finished with one storyboard, go on to the next piece of the consolidated vision that needs to be storyboarded. You will create a storyboard for each of your major tasks that are affected by the vision. You may have four to ten storyboards depending on the complexity of your problem, each with its own branches. Your goal is to cover the key tasks that the users will engage in to get their work done that are touched by introducing your new technology.